

CLINIC NAME

Clinic Management, Doctor, Staff & Patient Mobile Application

Project Proposal

1. Project Overview

CLINIC NAME will have a single Flutter mobile application designed to manage the complete clinic workflow from patient registration and OP management to doctor consultation, appointments, prescriptions, billing, pharmacy inventory, staff attendance and patient follow-up.

The application will support:

- Doctor Login
- Staff Login
- Customer/Patient Login
- Multiple clinic/hospital locations
- Appointment management
- Patient history
- Doctor availability calendar
- OP registration
- Vitals recording
- Case notes
- Prescription management
- Consultation fee collection
- Billing and collection reports
- Pharmacy inventory
- Staff attendance
- Follow-up reminders
- Push notifications
- Quick clinic contact
- Firebase backend

The application will be designed so that a small clinic can operate with one location, while the same system can support up to approximately 5 locations/branches.

2. Application Concept

The application will use a **single Flutter application**.

When the application opens, users will first see the available hospitals/clinic locations.

There will be no complicated hospital → branch selection process.

For example:

Select Hospital / Clinic

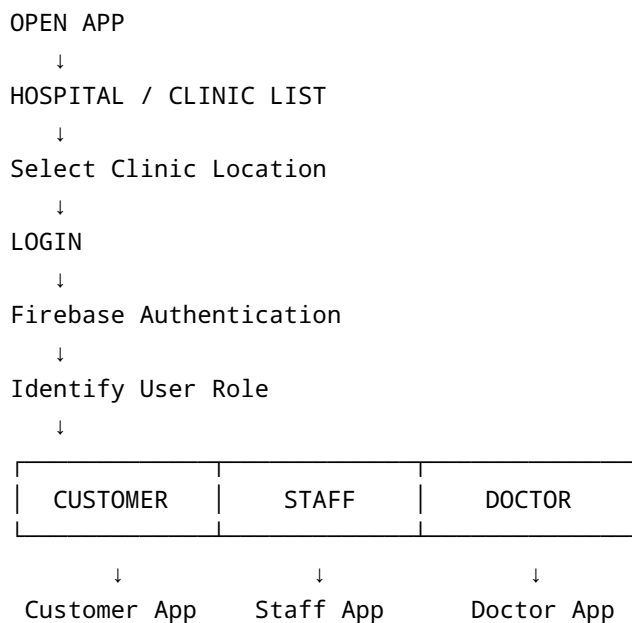
ABC Healthcare – Trichy Main Branch

ABC Healthcare – Srirangam Branch

ABC Healthcare – Thillainagar Branch

The user simply selects the required clinic/location and proceeds to login.

3. Main Application Flow



The user does not need to manually select Doctor/Staff/Customer after login if the account is already assigned a role.

4. Clinic / Hospital Location Management

The system will support multiple clinic locations.

Example:

CLINIC NAME

1. CLINIC NAME – Trichy Main Branch

2. CLINIC NAME - Srirangam Branch
3. CLINIC NAME - Thillainagar Branch
4. CLINIC NAME - Woraiyur Branch
5. CLINIC NAME - Manachanallur Branch

The maximum number of locations can be controlled from the administration system.

Each location can have its own:

- Address
- Phone number
- WhatsApp number
- Working hours
- Doctors
- Staff
- Appointments
- Patient activity
- Pharmacy stock
- Collection details

5. Login System

The application will have a common login system.

Customer Login

Customer can login using:

- Mobile number
- OTP authentication

Staff Login

Staff can login using:

- Mobile/email
- Password or secure authentication

Doctor Login

Doctor can login using:

- Mobile/email
- Password or secure authentication

Firebase Authentication will be used for secure authentication.

6. Customer / Patient Module

The customer application will provide patients with access to their clinic-related information.

Customer Dashboard

The dashboard will display:

- Patient name
- Patient ID
- Upcoming appointment
- Doctor name
- Appointment date/time
- Quick appointment booking
- Patient history
- Prescriptions
- Follow-up information
- Contact clinic

Example:

```
Welcome, Patient Name

Patient ID: PAT-000123

Upcoming Appointment
Dr. Doctor Name
15 Aug 2026
10:30 AM

[ View Appointment ]

[ Book Appointment ]

[ My History ]

[ Prescriptions ]

[ Follow-up ]

📞 Call Clinic
```

7. Existing Patient Linking

If a patient already exists in the clinic database, the customer should not receive a duplicate patient record.

The staff may have already created:

Patient ID: PAT-000123
Mobile: XXXXXXXX

When the patient later registers in the application using the same verified mobile number, the system can link the customer account to the existing patient record.

This allows the customer to view their permitted historical information.

8. New Patient Registration

For a new patient, the system can collect:

- Patient name
- Mobile number
- Gender
- Date of birth / age
- Address
- Basic profile information

The system will automatically generate a unique Patient ID.

Example:

PAT-000001
PAT-000002
PAT-000003

The Patient ID format can be configurable.

Example:

CLINIC-PAT-000001

9. Customer Doctor Availability

Customers can see available doctors and their available appointment slots.

Example:

Dr. Raj
General Medicine

Monday
09:00 AM - 01:00 PM

Wednesday
09:00 AM - 01:00 PM

Friday
04:00 PM - 08:00 PM

Customers can select:

1. Doctor
2. Date
3. Available time
4. Appointment reason
5. Submit appointment request

Only available slots will be displayed.

Already booked slots will not be available for selection.

10. Customer Appointment Request

Customer selects:

Doctor
Dr. Raj

Date
18 Aug 2026

Time
10:30 AM

Then submits the appointment request.

Appointment status:

- Requested
- Confirmed
- Checked In
- In Consultation
- Completed
- Cancelled

- Rescheduled
- No Show

The customer receives a notification when the appointment is confirmed or changed.

11. Staff OP Management

Staff can create the patient's OP entry.

Existing Patient

Staff can search using:

- Phone number
- Patient ID
- Patient name

If found, the existing patient record will open.

New Patient

Staff can create a new patient.

12. OP Patient Entry

For each visit, staff can enter:

- Patient
- Date
- Time
- Doctor
- OP number
- Reason for visit
- BP
- Height
- Weight

Additional vital fields can be added later if required:

- Pulse
 - Temperature
 - SpO2
 - Blood sugar
-

13. Appointment Creation by Staff

Staff can create appointments for walk-in or phone-booked patients.

Staff can select:

- Doctor
- Date
- Time
- Appointment type
- Reason
- Patient

The appointment will immediately appear in the doctor's appointment queue.

14. Doctor Dashboard

Doctor dashboard will provide a quick overview.

Today's Appointments	18
Waiting Patients	4
In Consultation	1
Completed	13
Today's Collection	₹11,500
Follow-ups	5

Main sections:

- Today's Appointments
 - Patient Search
 - Patient History
 - Case Notes
 - Prescriptions
 - Follow-ups
 - Billing
 - Pharmacy
 - Staff Attendance
 - Reports
-

15. Doctor Appointment Screen

Doctor can open an appointment and see:

Patient Details

- Patient name
- Patient ID
- Age
- Gender
- Contact information

Today's Vitals

- BP
- Height
- Weight
- Other recorded vitals

Previous History

Doctor can view previous clinic visits and permitted historical information.

16. Doctor Case Notes

Doctor can enter:

- Chief complaint
- Clinical findings
- Diagnosis
- Treatment advice
- Case notes
- Additional instructions

Doctor notes will be separated from patient-visible information where necessary.

This allows the clinic to keep internal notes private while showing appropriate information to the patient.

17. Doctor Prescription

Prescription is optional.

Doctor can add multiple medicines.

Each prescription item may contain:

- Medicine name
- Dosage
- Morning
- Afternoon

- Night
- Duration
- Before/after food
- Instructions

Example:

Paracetamol 500mg
1 - 0 - 1
5 Days
After Food

The customer can view the prescription through the application.

A prescription PDF feature can also be added if required.

18. Follow-Up Management

Doctor can select a follow-up date.

Example:

Follow-up Date:
20 Aug 2026

Follow-up Reason:
Review after medication

The follow-up will be stored against the patient's visit.

The system can automatically send push notifications to the customer.

Example:

Follow-up Reminder

Your follow-up with Dr. Raj is scheduled for tomorrow at 10:30 AM.

Another notification can be sent on the follow-up date.

19. Push Notification System

Firebase Cloud Messaging will be used for customer notifications.

Notifications can include:

Appointment Requested

"Your appointment request has been received."

Appointment Confirmed

"Your appointment with Dr. Raj has been confirmed."

Appointment Reminder

"Your appointment is tomorrow at 10:30 AM."

Follow-Up Reminder

"Your follow-up appointment is tomorrow."

Appointment Rescheduled

"Your appointment time has been changed."

Appointment Cancelled

"Your appointment has been cancelled."

Prescription Available

"Your prescription is now available in the app."

20. Patient History

Customer can view permitted previous visits.

Example:

12 Aug 2026

Dr. Raj

BP: 120/80

Weight: 68 kg

Diagnosis

...

Prescription

2 Medicines

Follow-up
20 Aug 2026

Older visits can also be displayed.

Doctor will have a more detailed clinical history based on their permission.

21. Customer Appointment History

Customer can see:

Upcoming

- Date
- Time
- Doctor
- Location
- Status

Completed

- Previous appointment
- Doctor
- Visit date
- Available medical information

Cancelled

- Appointment
 - Date
 - Cancellation status
-

22. Quick Contact

A permanent quick contact section/button will be available.

Call Clinic

Customer can tap the button to directly call the selected clinic location.

Optional additional buttons:

- WhatsApp
- Get Directions
- Clinic Address

For multi-location clinics, the phone number and address will belong to the selected clinic location.

23. Doctor Calendar Management

Doctors can configure their availability.

Example:

Monday
09:00 AM - 01:00 PM

Tuesday
04:00 PM - 08:00 PM

Wednesday
09:00 AM - 01:00 PM

Settings can include:

- Working days
- Start time
- End time
- Break time
- Appointment slot duration
- Maximum appointments
- Leave dates
- Holiday dates
- Branch/location

Example:

Slot Duration: 15 Minutes

09:00
09:15
09:30
09:45
10:00

Customers will only see valid available slots.

24. Multi-Location Doctor Schedule

A doctor can work at more than one location.

Example:

Dr. Raj

Trichy Main Branch
Monday / Wednesday / Friday

Srirangam Branch
Tuesday / Thursday

The system will show customers the appropriate doctor availability for each clinic location.

25. Consultation Fee Management

Fee collection can be performed by either:

Staff

Example:

Consultation Fee: ₹500
Collected: ₹500
Collected By: Staff
Payment: Cash

Doctor

Example:

Consultation Fee: ₹500
Collected: ₹500
Collected By: Doctor
Payment: UPI

The payment record will store who collected the amount.

26. Partial Payment

The system can support partial payment.

Example:

Consultation Fee	₹500
Paid	₹300
Balance	₹200

This provides flexibility for clinics that allow later payment.

27. Billing / Collection Report

Billing reports will be restricted to authorized Doctor/Admin users.

Today's Collection

Staff Collection	₹8,000
Doctor Collection	₹4,500

Total	₹12,500

Payment method:

Cash	₹5,000
UPI	₹7,000
Card	₹500

28. Monthly Collection

Doctor/Admin can view:

- Total appointments
- Completed appointments
- Total consultation fees
- Staff collection
- Doctor collection
- Cash collection
- UPI collection
- Card collection
- Outstanding amount

Date filters:

- Today
- Yesterday
- This Week
- This Month

- Custom Date Range
-

29. Pharmacy Inventory

The application will include pharmacy inventory management.

Medicine Master

- Medicine name
 - Generic name
 - Brand
 - Batch
 - Expiry date
 - Purchase price
 - Selling price
 - Current quantity
 - Minimum stock quantity
-

30. Pharmacy Stock IN

When medicines are purchased:

```
Medicine
Paracetamol 500mg

Batch
P1234

Quantity
500

Expiry
12/2027
```

Stock automatically increases.

31. Pharmacy Stock OUT

When medicine is dispensed or used:

```
Medicine
Paracetamol 500mg
```

Quantity

10

Reason

Patient Dispensing

Stock automatically decreases.

Prescription creation alone should not necessarily reduce stock. Stock should reduce when the medicine is actually dispensed/used.

32. Pharmacy Reports

Doctor/Admin can view:

- Current stock
- Low stock
- Stock IN
- Stock OUT
- Expired medicines
- Expiring medicines
- Batch details
- Stock movement
- Medicine usage

Example:

Opening Stock	500
Stock IN	300
Stock OUT	120
Adjustment	-5

Current Stock	675

33. Low Stock Alert

When stock falls below the configured minimum:

LOW STOCK

Paracetamol 500mg

Available: 15

Minimum: 25

The doctor/admin can see this on the dashboard.

34. Staff Attendance

Doctor/Admin can manage staff attendance.

Attendance can include:

- Present
- Absent
- Leave
- Check-in
- Check-out
- Notes

Monthly report:

Staff Name	Present	Absent	Leave
Staff 1	24	2	1
Staff 2	25	1	1

Staff self check-in/check-out can be supported if required.

35. Role-Based Permissions

Module	Doctor	Staff	Customer
Login	✓	✓	✓
Patient Registration	✓	✓	Limited
Patient Search	✓	✓	Own
OP Entry	✓	✓	—
Vitals	✓	✓	View
Appointment Creation	✓	✓	Request
Appointment View	✓	✓	Own
Doctor Calendar	✓	✓	View
Case Notes	✓	—	Selected info
Prescription	✓	—	View
Follow-up	✓	—	View

Module	Doctor	Staff	Customer
Fee Collection	✓	✓	View own
Billing Reports	✓	—	—
Pharmacy	✓	Optional	—
Pharmacy Reports	✓	—	—
Staff Attendance	✓	Own	—
Patient History	✓	Limited	Own
Notifications	✓	✓	✓

36. Firebase Backend

The application will use Firebase services.

Firestore Authentication

Used for:

- Customer login
- Staff login
- Doctor login
- OTP authentication
- Secure account management

Cloud Firestore

Used for:

- Clinics
- Locations
- Users
- Patients
- Appointments
- Visits
- Vitals
- Case notes
- Prescriptions
- Payments
- Follow-ups
- Doctor schedules
- Pharmacy
- Attendance
- Notifications

Firestore Cloud Messaging

Used for:

- Appointment notifications
- Follow-up reminders
- Confirmation
- Rescheduling
- Prescription notifications

Firestore Cloud Functions

Used for automated backend processes such as:

- Follow-up notification scheduling
- Appointment reminders
- Notification processing
- Data validation
- Automated status updates

Firestore Storage

Used for:

- Prescription files
- Patient documents if required
- Clinic images/logos
- Other authorized documents

Firestore App Check

Used as an additional application security layer.

37. Suggested Firestore Data Structure

```
clinics
└─ clinicId

locations
└─ locationId

users
└─ uid

patients
└─ patientId

appointments
└─ appointmentId
```

```
visits
  └─ visitId

vitals
  └─ vitalId

caseNotes
  └─ noteId

prescriptions
  └─ prescriptionId

prescriptionItems
  └─ itemId

payments
  └─ paymentId

followUps
  └─ followUpId

doctors
  └─ doctorId

doctorSchedules
  └─ scheduleId

medicines
  └─ medicineId

medicineBatches
  └─ batchId

stockTransactions
  └─ transactionId

staffAttendance
  └─ attendanceId

notifications
  └─ notificationId
```

Important records will contain appropriate:

```
clinicId
locationId
createdBy
```

```
createdAt
updatedAt
```

This ensures data separation between clinic locations.

38. Security

Medical and financial information must be protected.

The application will use role-based access control.

Customer

Can access only their own permitted information.

Staff

Can access only the clinic/location information required for their work.

Doctor

Can access authorized patient clinical information, appointments, billing and other permitted modules.

Admin

Can manage the complete clinic/location system.

Firebase Authentication and Firestore Security Rules will be used to enforce access at the backend level.

Permissions will not depend only on hiding buttons inside the Flutter application.

39. Admin Management

A web-based Admin Panel is recommended for managing the application.

Admin can manage:

Clinic

- Clinic name
- Logo
- Address
- Phone
- WhatsApp
- Working hours

Locations

- Location name
- Address
- Phone
- Map location
- Working hours

Doctors

- Doctor profile
- Specialization
- Branch/location
- Availability
- Consultation fee

Staff

- Staff profile
- Assigned location
- Permissions
- Attendance

Patients

- Patient details
- Patient ID
- History
- Appointments

Pharmacy

- Medicines
- Batches
- Stock
- Expiry

Reports

- Appointments
- Collections
- Pharmacy
- Attendance

40. Patient ID Configuration

Admin can configure the patient ID format.

Example:

Prefix:
PAT

Starting Number:
100001

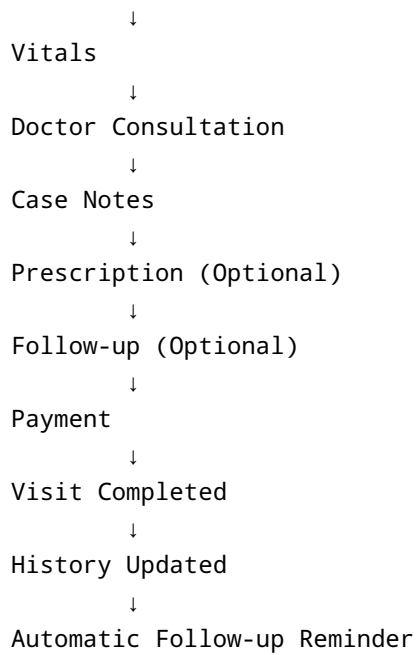
Result:

PAT-100001
PAT-100002
PAT-100003

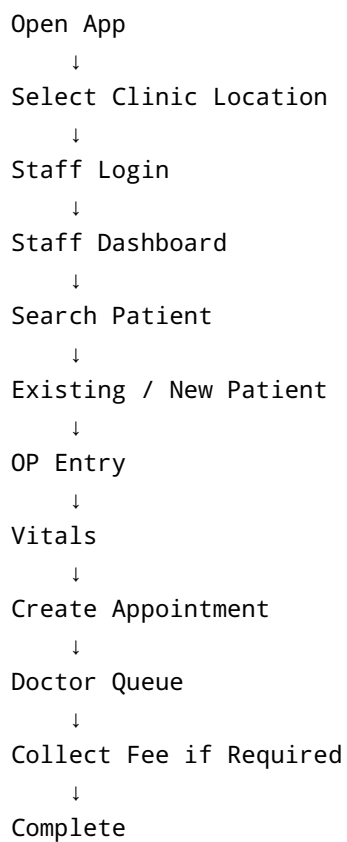
The ID should preferably remain unique across the entire clinic organization so that the same patient can visit multiple locations without receiving duplicate IDs.

41. Complete Patient Journey

```
Customer Opens App
↓
Selects Clinic Location
↓
Customer Login
↓
Existing Patient Linked
OR
New Patient Profile
↓
Customer Dashboard
↓
View Doctors
↓
View Availability
↓
Select Date
↓
Select Time
↓
Appointment Request
↓
Staff/Doctor Confirms
↓
Customer Notification
↓
Patient Visits Clinic
↓
Staff OP Entry
```



42. Complete Staff Journey



43. Complete Doctor Journey

```
graph TD; A[Open App] --> B[Select Clinic Location]; B --> C[Doctor Login]; C --> D[Doctor Dashboard]; D --> E[Today's Appointment Queue]; E --> F[Open Patient]; F --> G[View Previous History]; G --> H[View Today's Vitals]; H --> I[Case Notes]; I --> J[Diagnosis]; J --> K[Prescription]; K --> L[Follow-up]; L --> M[Fee Collection if Doctor Collected]; M --> N[Complete Visit];
```

44. Customer Notification Journey

```
graph TD; A[Appointment Requested] --> B[Notification]; B --> C[Appointment Confirmed]; C --> D[Notification]; D --> E[Appointment Tomorrow]; E --> F[Reminder]; F --> G[Appointment Completed];
```

↓
History Updated

Follow-up Date Set

↓
Automatic Reminder

Prescription Added

↓
Notification

45. Application Design

The Flutter application will use a clean and modern medical/clinic interface.

Design priorities:

- Simple navigation
- Large readable buttons
- Clean cards
- Clear appointment status
- Easy patient search
- Fast OP entry
- Minimal steps for staff
- Easy doctor consultation screen
- Simple customer dashboard
- Responsive layouts
- Android and iOS support

The first screen will focus on the clinic/location cards rather than a complicated login process.

46. Main Navigation

Customer

Home
Appointments
History
Profile

Staff

Dashboard
Patients

Appointments
OP
Profile

Doctor

Dashboard
Appointments
Patients
Reports
More

Modules such as Pharmacy, Attendance and Settings can be placed under the appropriate More/Management section.

47. Future Expansion Possibilities

The architecture will allow future features such as:

- Online consultation
- Video consultation
- Online payment
- Pharmacy billing
- Lab reports
- Medical documents
- Prescription PDF
- WhatsApp integration
- SMS integration
- Email notifications
- Queue/token management
- Patient feedback
- Doctor ratings
- Multiple doctors
- Multiple specialties
- Insurance information
- Expense management
- Advanced analytics

These can be added later without changing the basic application structure.

48. Recommended Technology

Mobile Application

Flutter

One codebase for:

- Android
- iOS

Backend

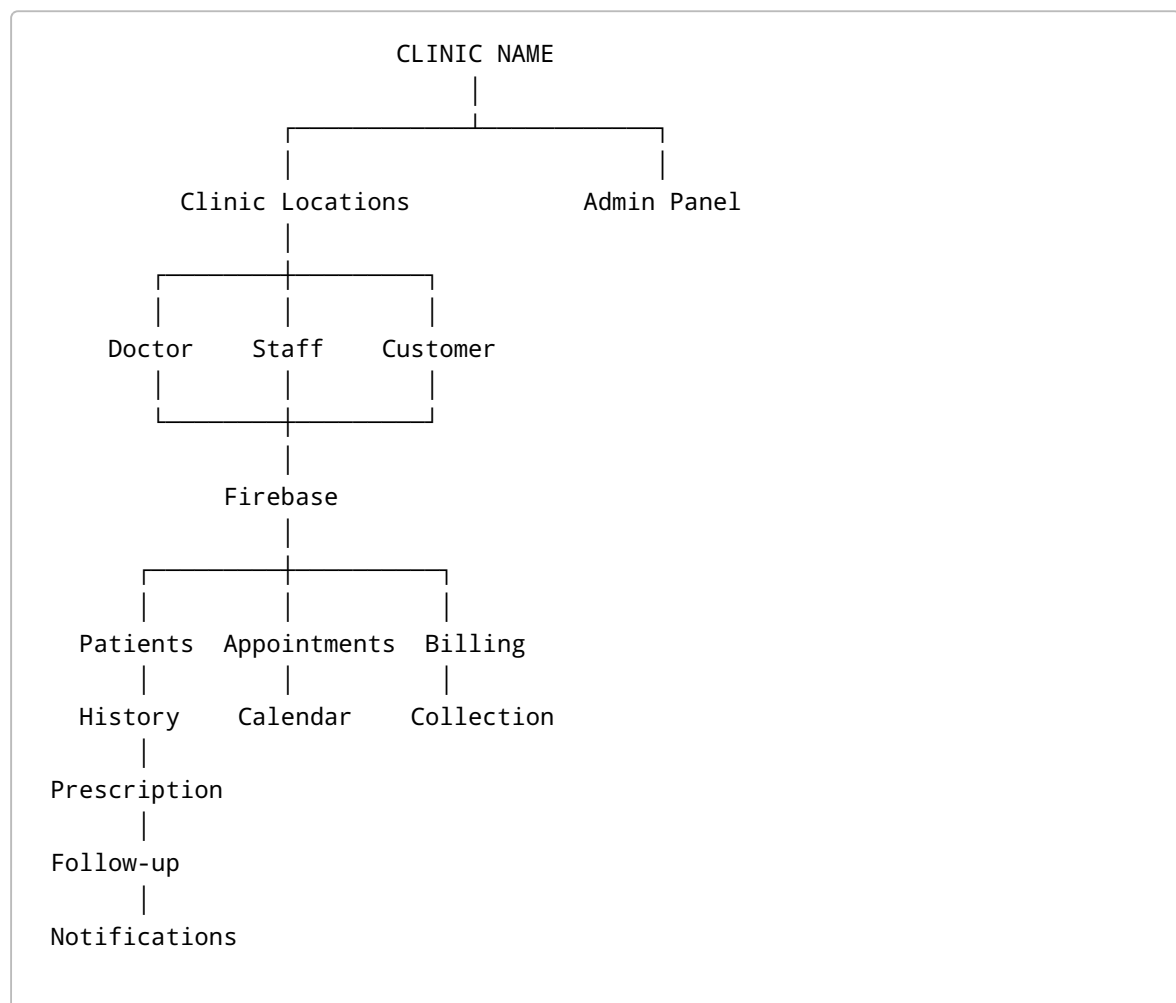
Firestore

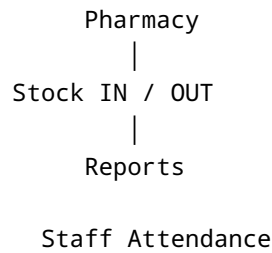
- Firebase Authentication
- Cloud Firestore
- Cloud Functions
- Firebase Cloud Messaging
- Firebase Storage
- Firebase App Check

Admin

Web-based Admin Panel connected to the same Firebase backend.

49. Final System Structure





50. Project Outcome

The final system will provide **CLINIC NAME** with a single integrated application for:

- Patient management
- OP management
- Doctor consultation
- Staff operations
- Customer appointment booking
- Doctor availability
- Patient history
- Prescription management
- Follow-up reminders
- Billing and collection
- Pharmacy inventory
- Staff attendance
- Multi-location management
- Push notifications
- Secure Firebase backend

The system is intentionally designed to remain **simple for small clinics**, while supporting approximately **1-5 clinic locations** and allowing future expansion.

Primary User Flow

Open App → Select Hospital/Clinic → Login → Role-based Dashboard → Perform Activity

This keeps the application easy for patients, staff and doctors while maintaining a scalable backend architecture.